Practice Set B

Class 12 Mathematics, Chapter 1: Relations and Functions

Maximum marks: **52** **20** questions

Name _____ Date _____ Score _____ / 52

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1 — 8 marks

1 MCQ, NCERT Ex 1.1 Q15 — 1 mark

Let R be the relation in the set $\{1, 2, 3, 4\}$ given by $R = \{(1, 2), (2, 2), (1, 1), (4, 4), (1, 3), (3, 3), (3, 2)\}$. Then:

(A) R is reflexive and symmetric but not transitive

(B) R is reflexive and transitive but not symmetric

(C) R is symmetric and transitive but not reflexive

(D) R is an equivalence relation

2 MCQ, NCERT Misc. Ex. Q6 — 1 mark

Let $A = \{1, 2, 3\}$. Then the number of relations containing $(1, 2)$ and $(1, 3)$ which are reflexive and symmetric but not transitive is:

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

3 MCQ, PYQ-style — 1 mark

Let $f : \mathbb{R}_+ \to [-5, \infty)$ be defined as $f(x) = 9x^2 + 6x - 5$, where $\mathbb{R}_+$ is the set of all non-negative real numbers. Then f is:

(A) one-one but not onto **(B)** onto but not one-one

(C) bijective **(D)** neither one-one nor onto

4 MCQ, CBSE 2024 (65/3/3) Q8 — 1 mark

Let $\mathbb{Z}$ denote the set of integers, then function $f : \mathbb{Z} \to \mathbb{Z}$ defined as $f(x) = x^3 - 1$ is:

(A) both one-one and onto **(B)** one-one but not onto

(C) onto but not one-one **(D)** neither one-one nor onto

5 MCQ, NCERT Exemplar Ex 1.3 Q35 — 1 mark

If the set A contains 5 elements and the set B contains 6 elements, then the number of one-one and onto mappings from A to B is:

(A) 720 **(B)** 120

(C) 0 **(D)** none of these

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** The relation $R = \{(a, b) : ab > 0\}$ on the set $\mathbb{R}$ of real numbers is symmetric and transitive but not reflexive. **Reason (R):** Every relation which is symmetric and transitive is also reflexive. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, Board-pattern Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 4x^3 + 5$ is both one-one and onto.	2 marks

Level 2

19 marks

8	Short answer, Board-pattern Show that the relation R on the set $\mathbb{Z}$ of integers, given by $R = \{(a, b) : a^2 + b^2 \text{ is divisible by } 2\}$, is an equivalence relation.	2 marks
9	Short answer, CBSE 2024 (65/4/1) Q33(b) Check whether the relation S in the set of real numbers $\mathbb{R}$ defined by $S = \{(a, b) : a - b + \sqrt{2} \text{ is an irrational number}\}$ is reflexive, symmetric or transitive.	3 marks
10	Short answer, Board-pattern Let $A = \{1, 2, 3, \ldots, 12\}$ and R be the relation in $A \times A$ defined by $(a, b)\, R\, (c, d)$ if $ab = cd$. Prove that R is an equivalence relation and obtain the equivalence class $[(3, 4)]$.	3 marks
11	Short answer, Board-pattern Let $f : \mathbb{N} \to \mathbb{Z}$ be defined by $f(n) = \begin{cases} \frac{n-1}{2}, & \text{if } n \text{ is odd} \\ -\frac{n}{2}, & \text{if } n \text{ is even} \end{cases}$. Show that f is both one-one and onto.	3 marks
12	Short answer, Board-pattern Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \frac{x^2}{1 + x^2}$ is neither one-one nor onto. Further, find the set A so that $f : \mathbb{R} \to A$ becomes an onto function.	3 marks
13	Short answer, PYQ-style Let $A = \mathbb{R} - \{-1\}$ and $f : A \to \mathbb{R} - \{b\}$ be defined by $f(x) = \frac{2x - 3}{x + 1}$. Find the value of b for which f is onto. For this b, show that f is also one-one.	3 marks
14	Short answer, NCERT Exemplar Ex 1.3 Q17 Let R be the relation defined on the set of natural numbers $\mathbb{N}$ as $R = \{(x, y) : x \in \mathbb{N}, y \in \mathbb{N}, 2x + y = 41\}$. Find the domain and range of the relation R. Also verify whether R is reflexive, symmetric and transitive.	2 marks

Q No.	Question	Marks

Level 3 — 25 marks

15 — Short answer, NCERT Ex 1.1 Q11 — 3 marks

Show that the relation R in the set A of points in a plane given by $R = \{(P, Q) : \text{distance of the point } P \text{ from the origin is same as the distance of the point } Q \text{ from the origin}\}$, is an equivalence relation. Further, show that the set of all points related to a point $P \neq (0, 0)$ is the circle passing through P with origin as centre.

16 — Long answer, CBSE SQP 2026-27 Q32(A) — 5 marks

Let $A = \mathbb{N} \times \mathbb{N}$, where $\mathbb{N}$ is the set of natural numbers and a relation R is defined on A such that $(a, b)\, R\, (c, d)$ if and only if $a^2 + d^2 = b^2 + c^2$. Check whether R is an equivalence relation or not.

17 — Long answer, NCERT Exemplar Ex 1.3 Q21 — 5 marks

Let $A = [-1, 1]$. Discuss whether the following functions from A to A are one-one, onto or bijective:

(i) $f(x) = \dfrac{x}{2}$ — 1 mark

(ii) $g(x) = |x|$ — 1 mark

(iii) $h(x) = x|x|$ — 2 marks

(iv) $k(x) = x^2$ — 1 mark

18 — Long answer, Board-pattern — 5 marks

Let $A = \{x \in \mathbb{Z} : 0 \leq x \leq 10\}$. Show that the relation $R = \{(a, b) : a, b \in A,\ a^2 - b^2 \text{ is divisible by } 5\}$ is an equivalence relation on A. Write the equivalence classes $[1]$ and $[2]$. How many distinct equivalence classes does R have?

19 — Short answer, Board-pattern — 3 marks

Show that the function $f : [1, 3] \to \mathbb{R}$ defined by $f(x) = x^2 - 2x + 3$ is one-one. Also find the range of f.

Q No.	Question	Marks
20	Case study, CBSE SQP 2024-25 Q37 **Case Study:** An organization conducted bike race under 2 different categories — boys and girls. In all, there were 250 participants. Among all of them finally three from Category 1 and two from Category 2 were selected for the final race. Ravi forms two sets B and G with these participants for his college project. Let $B = \{b_1, b_2, b_3\}$, $G = \{g_1, g_2\}$, where B represents the set of boys selected and G the set of girls who were selected for the final race. Ravi decides to explore these sets for various types of relations and functions.	4 marks
	(i) Ravi wishes to form all the relations possible from B to G. How many such relations are possible?	1 mark
	(ii) Write the smallest equivalence relation on G.	1 mark
	(iii)(a) Ravi defines a relation from B to B as $R_1 = \{(b_1, b_2), (b_2, b_1)\}$. Write the minimum ordered pairs to be added in R_1 so that it becomes (A) reflexive but not symmetric, (B) reflexive and symmetric but not transitive.	2 marks
	OR	
	(iii)(b) If the track of the final race (for the biker b_1) follows the curve $x^2 = 4y$ (where $0 \le x \le 20\sqrt{2}$ and $0 \le y \le 200$), then state whether the track represents a one-one and onto function or not. (Justify)	2 marks

Answer key

Chapter 1: Relations and Functions, Practice Set B. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) R is reflexive and transitive but not symmetric
2	(A) 1
3	(C) bijective
4	(B) one-one but not onto
5	(C) 0
6	(C) A is true, but R is false
7	f is one-one and onto (bijective).

Level 2

8	R is reflexive, symmetric and transitive, hence an equivalence relation.
9	S is reflexive, but neither symmetric nor transitive.
10	R is an equivalence relation; $[(3,4)] = \{(1,12),(2,6),(3,4),(4,3),(6,2),(12,1)\}$
11	f is both one-one and onto (bijective).
12	Neither one-one nor onto; $A = [0,1)$.
13	$b = 2$; f is one-one and onto.
14	Domain $= \{1,2,\ldots,20\}$, range $= \{1,3,5,\ldots,39\}$; R is neither reflexive, nor symmetric, nor transitive.

Level 3

15	R is an equivalence relation; points related to P form the circle with centre O and radius OP.
16	R is reflexive, symmetric and transitive, hence an equivalence relation.
17	(i) one-one, not onto (ii) neither one-one nor onto (iii) bijective (iv) neither one-one nor onto
18	R is an equivalence relation; $[1] = \{1,4,6,9\}$, $[2] = \{2,3,7,8\}$; 3 classes.
19	f is one-one; range of $f = [2,6]$.
20	(i) 64 (ii) $\{(g_1,g_1),(g_2,g_2)\}$ (iii)(a) (A) $(b_1,b_1),(b_2,b_2),(b_3,b_3),(b_2,b_3)$; (B) $(b_1,b_1),(b_2,b_2),(b_3,b_3),(b_2,b_3),(b_3,b_2)$ (other valid choices accepted) (iii)(b) one-one and onto